

Jacobian Descent For Multi-Objective Optimization

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Joint work with Pierre Quinton

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github.com/TorchJD/torchjd



torchjd.org



arxiv.org/pdf/2406.16232

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Theory

Multi-objective optimization

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \mathbf{u} \preceq \mathbf{v} \\ \Leftrightarrow & \quad u_i \leq v_i \quad \forall i \in [m] \end{aligned}$$

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Examples:

$$\blacktriangleright \begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

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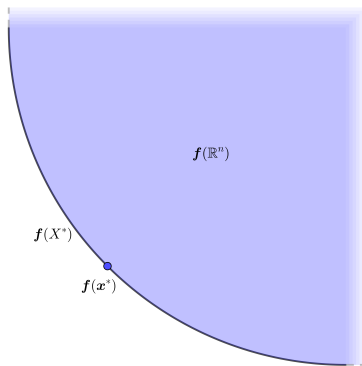
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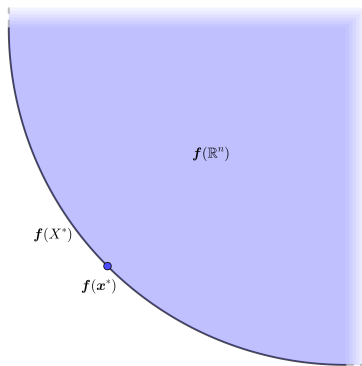
▶ $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



Pareto optimality

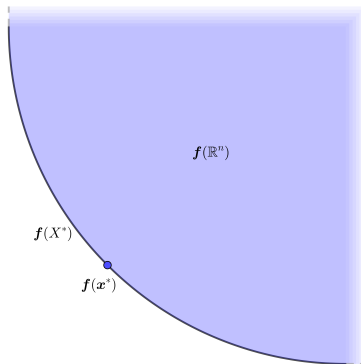
Pareto set: X^*



Pareto optimality

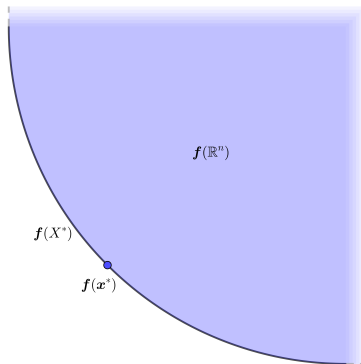
Pareto set: X^*

Pareto Front: $f(X^*)$



Pareto optimality

Pareto set: X^*
Pareto Front: $f(X^*)$



$$\mathbf{x}^* \in X^* \quad \Leftrightarrow \quad \forall \mathbf{y} \in \mathbb{R}^n, \mathbf{f}(\mathbf{y}) \preceq \mathbf{f}(\mathbf{x}^*) \rightarrow \mathbf{f}(\mathbf{y}) = \mathbf{f}(\mathbf{x}^*)$$

First-order Taylor approximation

For any $\mathbf{x} \in \mathbb{R}^n$, the Jacobian of $\mathbf{f}$ at $\mathbf{x}$, is

$$\mathcal{J}\mathbf{f}(\mathbf{x}) = \begin{bmatrix} \nabla f_1(\mathbf{x})^\top \\ \nabla f_2(\mathbf{x})^\top \\ \vdots \\ \nabla f_m(\mathbf{x})^\top \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} f_1(\mathbf{x}) & \frac{\partial}{\partial x_2} f_1(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_1(\mathbf{x}) \\ \frac{\partial}{\partial x_1} f_2(\mathbf{x}) & \frac{\partial}{\partial x_2} f_2(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_2(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} f_m(\mathbf{x}) & \frac{\partial}{\partial x_2} f_m(\mathbf{x}) & \cdots & \frac{\partial}{\partial x_n} f_m(\mathbf{x}) \end{bmatrix}$$

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For a small enough update $\mathbf{y} \in \mathbb{R}^n$, Taylor's theorem states

$$\mathbf{f}(\mathbf{x} + \mathbf{y}) \approx \mathbf{f}(\mathbf{x}) + \mathcal{J}\mathbf{f}(\mathbf{x}) \cdot \mathbf{y}$$

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We select $\mathbf{y} = -\eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x})) \in \mathbb{R}^n$.

Jacobian descent

The mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$ is called an **aggregator**, it parameterizes **Jacobian descent**:

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Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

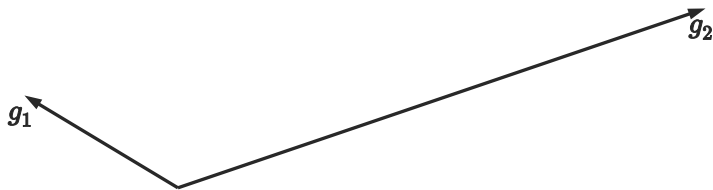
Input: $\mathbf{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^n$

for $t \leftarrow 1$ **to** T **do**

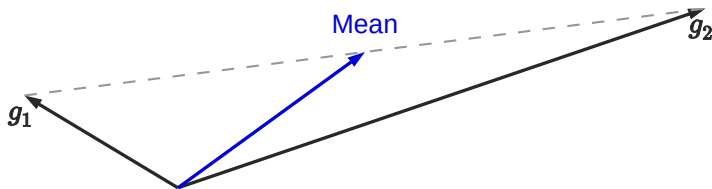
$\mathbf{x} \leftarrow \mathbf{x} - \eta \mathcal{A}(\mathcal{J}\mathbf{f}(\mathbf{x}))$

Output: $\mathbf{x}$

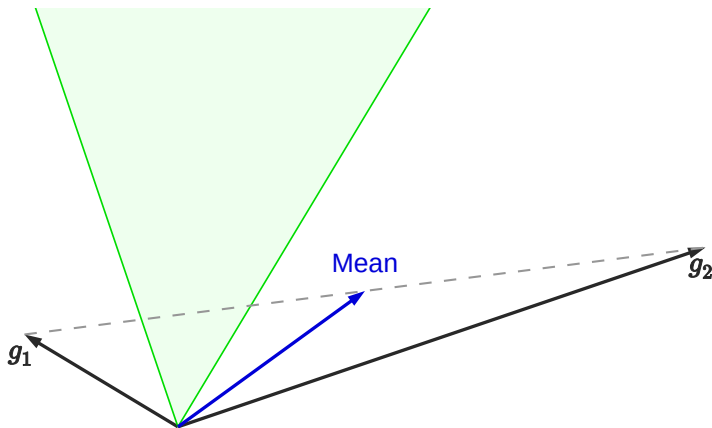
Unconflicting projection of gradients (UPGrad)



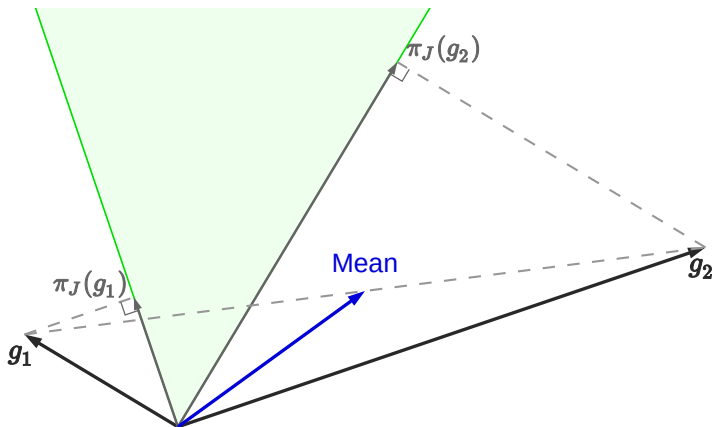
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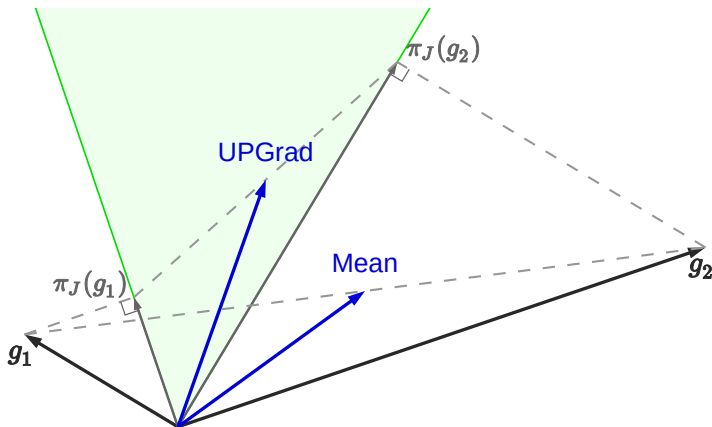
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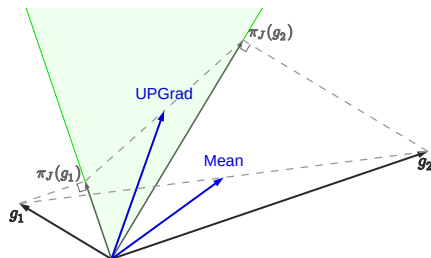
Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\mathbf{y} \in \mathbb{R}^n : \mathbf{0} \preceq J\mathbf{y}\}$. $\mathcal{A}$ is non-conflicting if for all $J \in \mathbb{R}^{m \times n}$, $\mathcal{A}(J) \in C_J$.

For any $\mathbf{x} \in \mathbb{R}^n$, the projection of $\mathbf{x}$ onto C_J is

$$\pi_J(\mathbf{x}) = \arg \min_{\mathbf{y} \in C_J} \|\mathbf{x} - \mathbf{y}\|^2$$

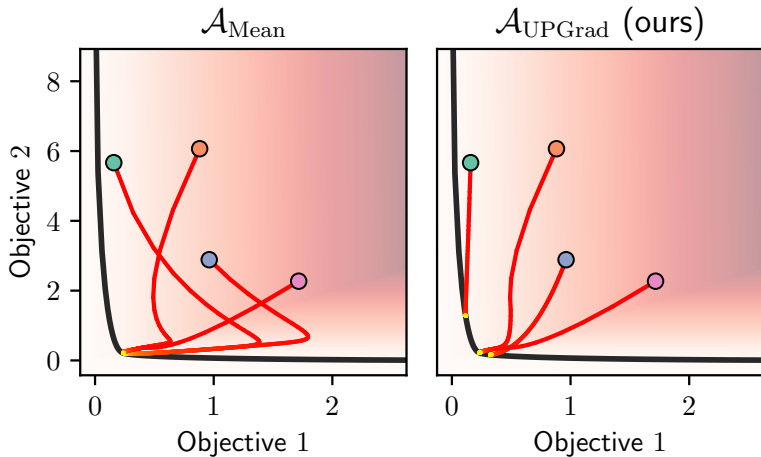
$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} \pi_J(J^\top \mathbf{e}_i)$$



Applications and experimentation

Optimization trajectories

Some convex quadratic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with conflicting gradients.



Optimization trajectories: code

```
x = tensor([0., 0.]) # initial value of x

for i in range(n_iters):
    y = f(x) # shape: [2]
    autojac.backward(y)
    # x now has a .jac field
    autojac.jac_to_grad([x], UPGrad())
    # x now has a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full project available at github.com/TorchJD/trajectories

Per-sample JD

Each element of the batch has its own loss.

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```
batch_size = 16
model = Linear(10, 1)

for x, y in dataloader:
    y_pred = model(x).squeeze(dim=1) # shape: [16]
    losses = loss_fn(y_pred, y) # shape: [16]
    autograd.backward(losses)
    # parameters now have a .grad field
    autograd.grad(model.parameters(), UPGrad())
    # parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/iwrm.

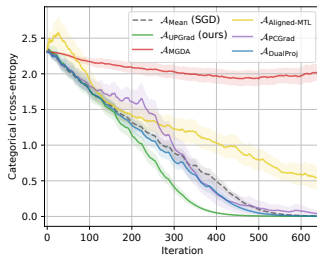
Per-sample JD: results

Batch size: 32

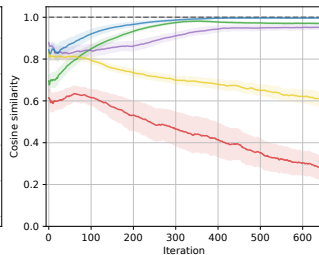
CIFAR-10



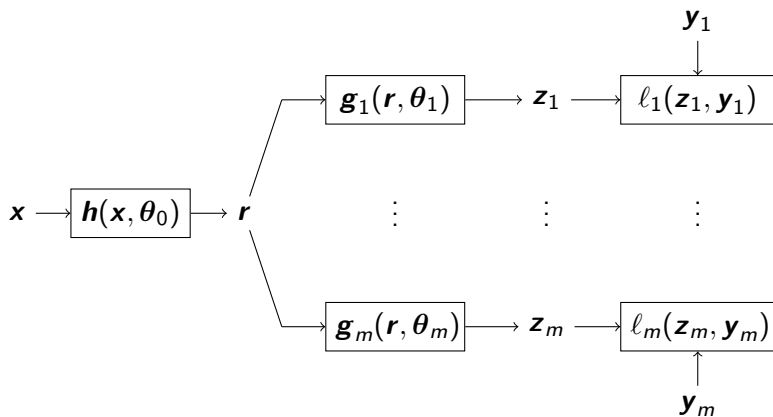
Training Loss



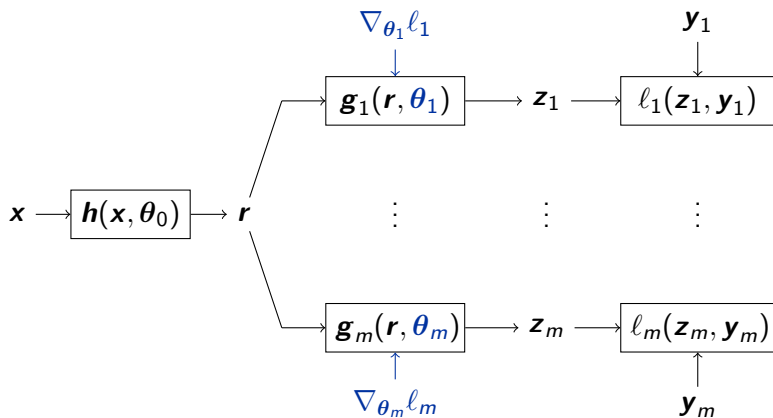
Update Similarity



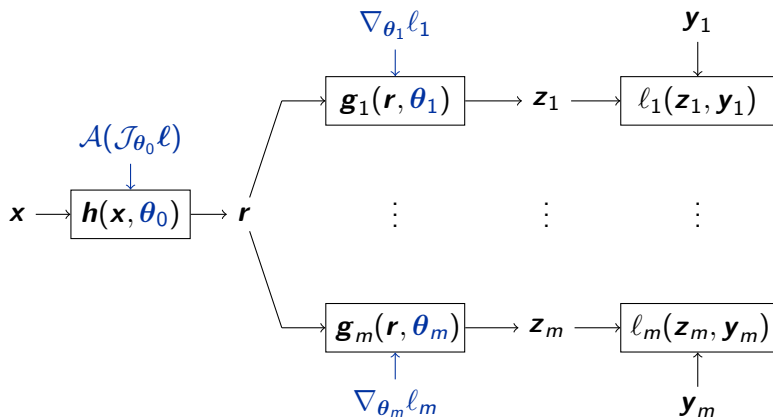
Multi-task learning



Multi-task learning



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Multi-task learning: code

```
backbone = Linear(10, 5)
head1 = Linear(5, 1)
head2 = Linear(5, 1)

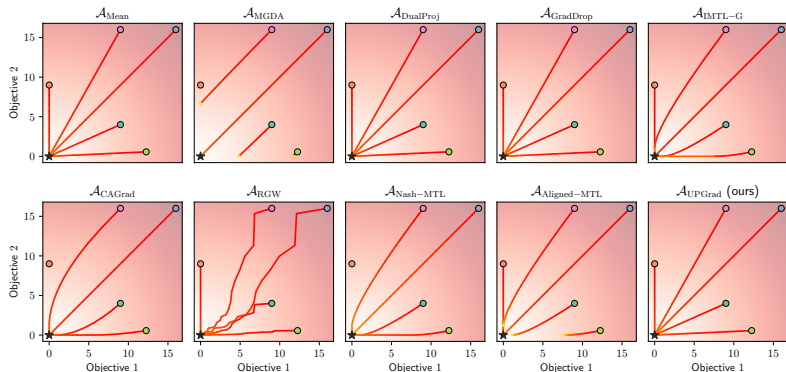
for x, y1, y2 in dataloader:
    features = backbone(x)
    y_pred1 = head1(features)
    y_pred2 = head2(features)
    loss1 = loss_fn(y_pred1, y1)
    loss2 = loss_fn(y_pred2, y2)
    autograd.backward([loss1, loss2], features)
    # backbone parameters now have a .jac field
    # and heads have a .grad field
    autograd.jac_to_grad(backbone.params(), UPGrad())
    # All parameters now have a .grad field
    optimizer.step()
    optimizer.zero_grad()
```

Full code example available at torchjd.org/latest/examples/mtl.

Conclusion

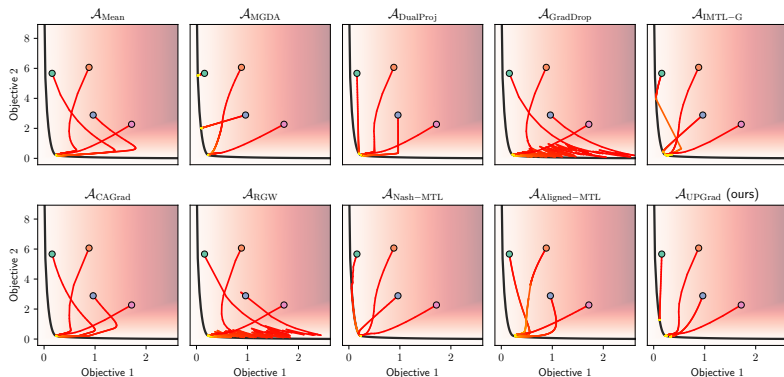
Bonus: Element-wise quadratic function trajectories

Minimize $\mathbf{f} : \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x^2 \\ y^2 \end{bmatrix}$. It's Jacobian at $\begin{bmatrix} x \\ y \end{bmatrix}$ is $\begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$.



Bonus: Convex quadratic form trajectories

Some convex quadratic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, with conflicting gradients.



Bonus: Gramian-based JD

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Applicable to most other aggregators from the literature.